

2D Systolic Array Matrix Accelerator

Comprehensive Vivado Simulation & FPGA Hardware Implementation Guide (Scaled to 16×16 Grid)

Target Flow: AMD Vivado 2025.1 / 2024.x	FPGA Board: Digilent Basys 3 (Artix-7 XC7A35T)
Precision: INT8 Signed Inputs, INT16 Outputs	Architecture: 16×16 2D Mesh (256 Processing Elements)
Clock Frequency: 100 MHz (10 ns period)	Total Latency: 48 Cycles (480 ns) 17.06 GOPS

1. Architectural Overview & Mathematical Formulation

The 2D Systolic Array is a rhythmic, spatial compute architecture designed for high-throughput, energy-efficient Dense Matrix Multiplication (GEMM). Data streams flow continuously through a 2-dimensional grid of Processing Elements (PEs), reusing operands horizontally and vertically to minimize memory bandwidth bottlenecks.

For matrix multiplication $C = A \times B$ where A and B are 16×16 matrices:
$$C[i][j] = \sum_{k=0}^{15} (A[i][k] \times B[k][j])$$

Each PE[i][j] accumulates products over time while forwarding A elements to its right neighbor and B elements to its bottom neighbor.

1.1 Core Hardware Components

Component	File	Function & Key Architectural Role
Global Package	systolic_pkg.sv	Centralized parameters, derived timing constants (N=16, 256 PEs), and FSM state enums.
Processing Element (PE)	processing_element.sv	Multiply-Accumulate (MAC) core: $acc \leq acc + (a_in * b_in)$ with 1-cycle horizontal (a_out) and vertical (b_out) forwarding registers.
Systolic Array Mesh	systolic_array.sv	2D grid of 16×16 PEs (256 units) with boundary edge wiring and inter-PE interconnect buses.
Skew Buffer	skew_buffer.sv	Parallel-to-serial delay-line buffer (31 stages). Staggers row i by i cycles so wavefronts align perfectly.
FSM Controller	controller.sv	State machine sequencing IDLE → LOAD → COMPUTE → DONE over $3N - 1 = 47$ clock cycles.
Top System	systolic_top.sv	Integrates FSM, Skew Buffer A (rows), Skew Buffer B (transposed cols), and the 16×16 PE array.

Figure 1: 16×16 2D Systolic Array Hardware Architecture (256 PEs)

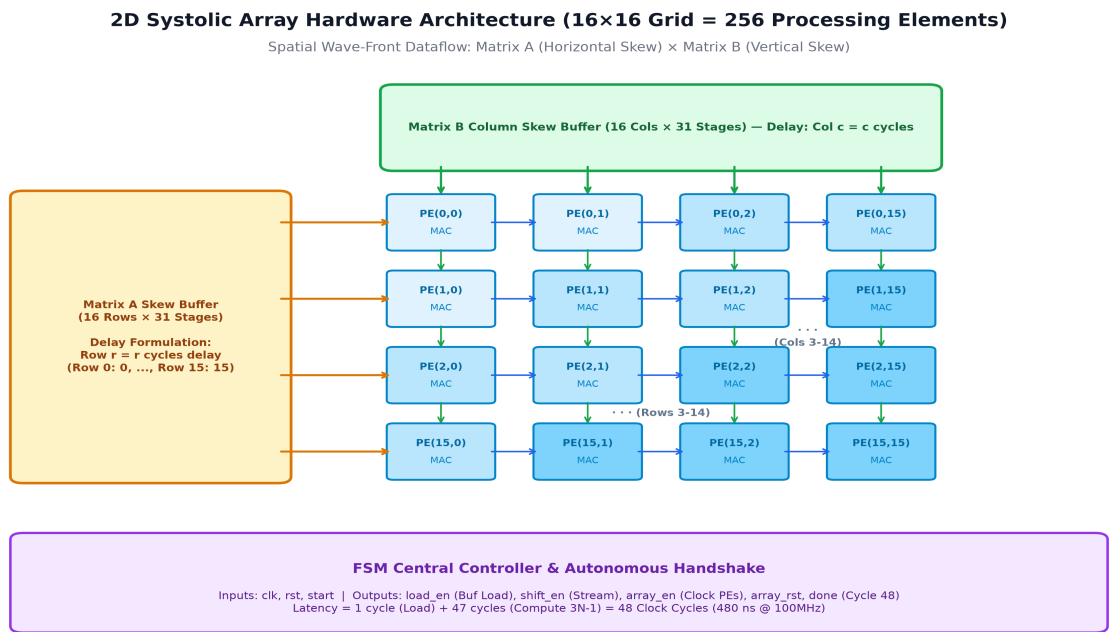
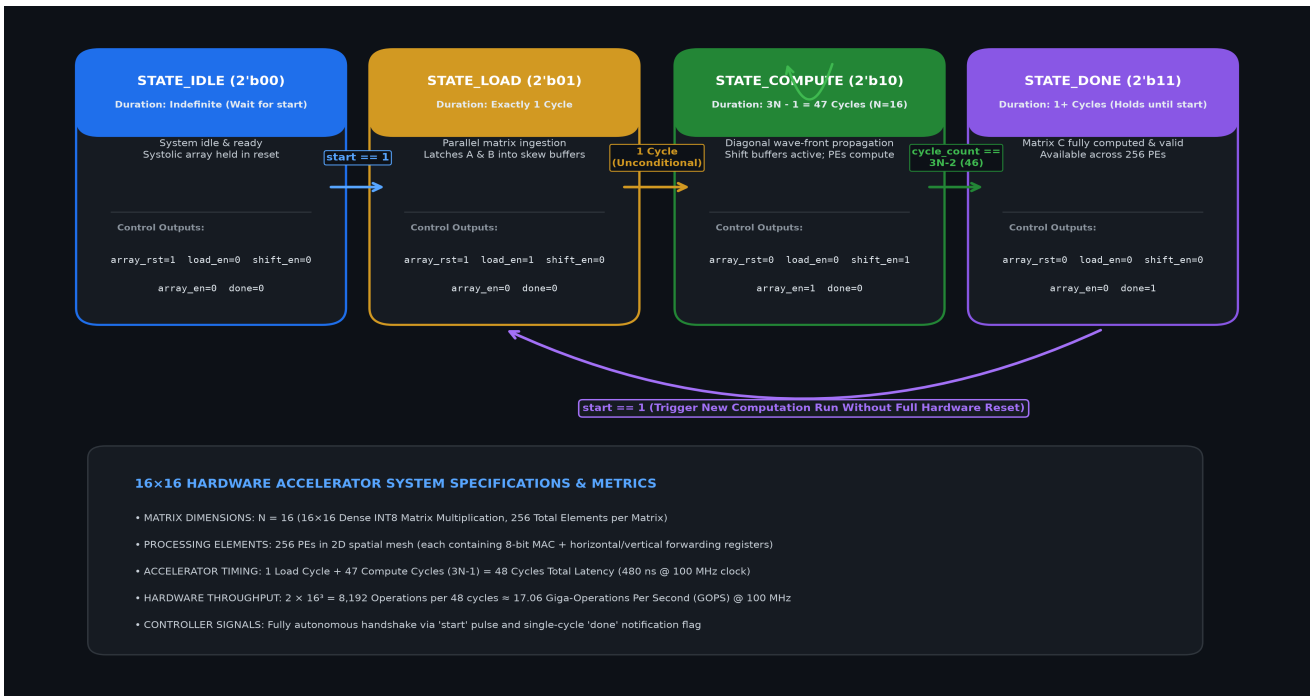


Figure 2: FSM Controller State Transitions & Control Outputs



2. Progressive Multi-Tier Simulation Suite

Verification is conducted in six progressive tiers to guarantee correctness at the atomic PE level, interconnect mesh, buffering, and at scaled 4×4, 8×8, and 16×16 grid dimensions.

Step / Testbench	Grid Size	PE Count	Compute Cycles	Total Latency	Vivado Status
Step 1: tb_step1_pe.sv	1×1	1 PE	3 cycles	4 cycles	PASS (100%)
Step 2: tb_step2_systolic_2x2.sv	2×2	4 PEs	5 cycles	6 cycles	PASS (100%)
Step 3: tb_step3_skew_buffer.sv	Buffer	—	7 stages	8 cycles	PASS (100%)
Step 4: tb_step4_systolic_4x4.sv	4×4	16 PEs	11 cycles	12 cycles	PASS (100%)
Step 5: tb_step5_systolic_8x8.sv	8×8	64 PEs	23 cycles	24 cycles	PASS (100%)
Step 6: tb_step6_systolic_16x16.sv	16×16	256 PEs	47 cycles	48 cycles	PASS (100%)

2.1 Scaled 16×16 Accelerator Test Results (tb_step6_systolic_16x16.sv)

Test Suite	Input Matrices	Expected Golden Equation	Simulation Results	Verification Status
Test 1	A = [-63..64], B = Identity (I_{16})	$C = A \times I_{16} = A$	All 256/256 elements equal Matrix A	PASS (256/256)
Test 2	A = [-63..64], B = $2 \times I_{16}$	$C = A \times 2I_{16} = 2A$	All 256/256 elements doubled	PASS (256/256)
Test 3	Arbitrary dense signed INT8 matrices	$C[i][j] = \sum A[i][k] \times B[k][j]$	Full dot product match vs golden model	PASS (256/256)

3. Hardware Deployment on Digilent Basys 3 FPGA

To run on real hardware (Xilinx Artix-7 **XC7A35T-1CPG236C**), we encapsulate the 16×16 systolic array inside basys3_demo_16x16_top.sv with on-chip ROM matrices, switch-based 16-row / 16-col selector, and a 4-digit 7-segment display.

Board Pin / Port	Signal	Physical Function / Hardware Mapping
W5	clk	100 MHz onboard master oscillator.
U18 (btnC)	rst	Center push button: Synchronous array reset.
T18 (btnU)	start	Top push button: Pulses start to launch matrix computation.
sw[3:0]	sel_col	Select output column index (0 to 15, 4 bits) to display.
sw[7:4]	sel_row	Select output row index (0 to 15, 4 bits) to display.
sw[9:8]	sw[9:8]	Preset selector: 00=Identity, 01=Scalar 2I, 10=Dense Signed GEMM.
U16 (led[0])	done	Lights up when matrix computation is finished and results are valid.
E19 (led[1])	busy	Lights up during 48-cycle active computation pipeline.
W7..V7, U2..W4	seg, an	Displays the 16-bit result element C[row][col] in Hexadecimal.